

VIKAS

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SUMMARY

I am a dedicated researcher with a strong passion for science. My interest is in studying proteins through molecular biology, biophysics, bioinformatics, and biochemistry. I believe in the value of interdisciplinary approach to understand and answer research questions. Currently, I am involved in projects exploring enzymatic engineering for improving plastic (PET) degradation. I have demonstrated a strong academic foundation by securing top ranks in highly competitive national examinations such as **All India Rank 21** in the CSIR-UGC NET (Life Sciences, 2025) and **All India Rank 75** in GATE XL (2026).

EDUCATION

BS-MS Dual Degree, Biology Major (Data Science Minor)

Nov 2021 – Jun 2026

Indian Institute of Science Education and Research (IISER) Mohali [\[website\]](#)

CPI: 8.0/10

Certifications

- Introduction to Antibody Engineering [\[certificate\]](#)
- Introduction to Biostatistics [\[certificate\]](#)

RESEARCH EXPERIENCE

Undergraduate and Master's Thesis

Dec 2023 – Present

Protein Design and Engineering Lab, IISER Mohali

I joined the Protein Design and Engineering Lab as an undergraduate intern to learn various molecular biology techniques. Over time, my interest in protein research deepened, leading me to continue in the same lab as a Master's thesis student. Over the course of time I have worked on various projects:

- Completed a project on protein-protein interactions within the cellulosome of *Clostridium thermocellum*, which has resulted in a manuscript currently out for submission (second author)
- Presently, focusing on improving plastic-degrading enzymes (LCC and MHETase) through rational protein engineering. The work integrates computational analyses (such as modeling and molecular dynamics simulations) with experimental validation to improve enzyme stability and activity.
- Bridging a collaboration with industry for development of a pipeline for AI driven protein engineering of plastic degrading enzymes.

Summer Intern

May 2023 – Jul 2023

Atmospheric Chemistry and Emissions Research Group, IISER Mohali

Worked with gas analyzers, performed instrument calibrations, and conducted QA/QC of chemical data. Learned atmospheric chemistry and gained proficiency in data analysis using MATLAB and Igor Pro. Conducted seasonal analysis of Delhi meteorological data (2022) to identify unique weather events.

Summer Intern

Apr 2022 – Jul 2022

Evolutionary Biology Lab, IISER Mohali

Conducted experiments on behavioral and selection traits in *Drosophila*. Analyzed survivability after infection and performed dissections of larval salivary glands and fly seminal vesicles, gaining hands-on experience in experimental design and data collection.

RESEARCH SKILLS

Experimental

- Molecular cloning and protein biochemistry: primer design, PCR, digestion, ligation, transformation, plasmid isolation, gel electrophoresis, SDS-PAGE, enzyme activity assay.
- Protein purification using affinity chromatography (Ni-NTA, Gel Filtration) and biophysical characterization using UV-Vis, fluorescence and circular dichroism (CD) spectroscopy, High performance liquid chromatography (HPLC), Microscale Thermophoresis (MST), and Differential Scanning Calorimetry (DSC).
- Experience in *E. coli* expression systems and basic *Drosophila* handling.

Computation and Bioinformatics

- Protein sequence and structure analysis, protein structure modelling, molecular docking, and MD simulations using Schrödinger Suite.
- Structure visualization and interaction mapping using PyMOL, UCSF Chimera, and JalView.
- Rational protein design through data-driven bioinformatics.

Software and Programming

- PyMOL, Schrödinger Suite, UCSF Chimera, JalView, SnapGene, Origin Pro, MATLAB, MS Office.
- Basic shell scripting (Bash) and proficient in **Python**.

Publications

- (Pre-print) Exploring differential interactional preferences of enzyme-bearing dockerins for cohesin domains in the *Clostridium thermocellum* cellulosome
Snehal Waghmare, **Vikas Yadav**, Saji Menon, Purnananda Guptasarma
[\[https://doi.org/10.64898/2026.01.19.700288\]](https://doi.org/10.64898/2026.01.19.700288)

ACHIEVEMENTS

- ★ **All India Rank 75 in GATE XL (Life Sciences)** Mar 2026
Graduate Aptitude Test in Engineering(GATE) is a national entrance exam for Master's and PhD programs in science and engineering disciplines across India; secured position within the top 0.05%
- ★ **All India Rank 123 in GATE BT (Biotechnology)** Mar 2026
Same national exam in the Biotechnology stream; secured position within the top 0.05% of candidates
- ★ **All India Rank 21 in CSIR-UGC NET (Life Sciences)** July 2025
National-level eligibility test for lectureship, PhD and award of Junior Research Fellowships in India, taken by more than 60,000 candidates annually; **obtained a percentile score of 99.953**
- ★ **All India Rank 158 in GATE XL (Life Sciences)** Feb 2025
secured position within the top 0.5%

LEADERSHIP AND VOLUNTEERING

Co-Convener, Biology Discussion Forum (IISER Mohali) Sep 2022 – Oct 2023

I have guided 300+ undergraduates in professional development and research interests. I also organized talks by esteemed researchers, quizzes, bird walks, treasure hunts, and student presentations. I have also coordinated lab demonstrations for 2000+ school students on IISER Mohali Foundation Day.

Volunteer, Insomnia Cultural Fest 2022
IISER Mohali Marketing and Event Management

EXTRA-CURRICULAR ACTIVITIES

Sports and E-Sports

Football, Athletics, Valorant

LETTER OF RECOMMENDATION

Please contact:

- Prof. Purnananda Guptasarma (guptasarma@iisermohali.ac.in)
- Prof. Shashi Bhushan Pandit (shashibp@iisermohali.ac.in)